Problem 6.1 (Theoretical)
Extended Kalman filter and non-linear system state estimation

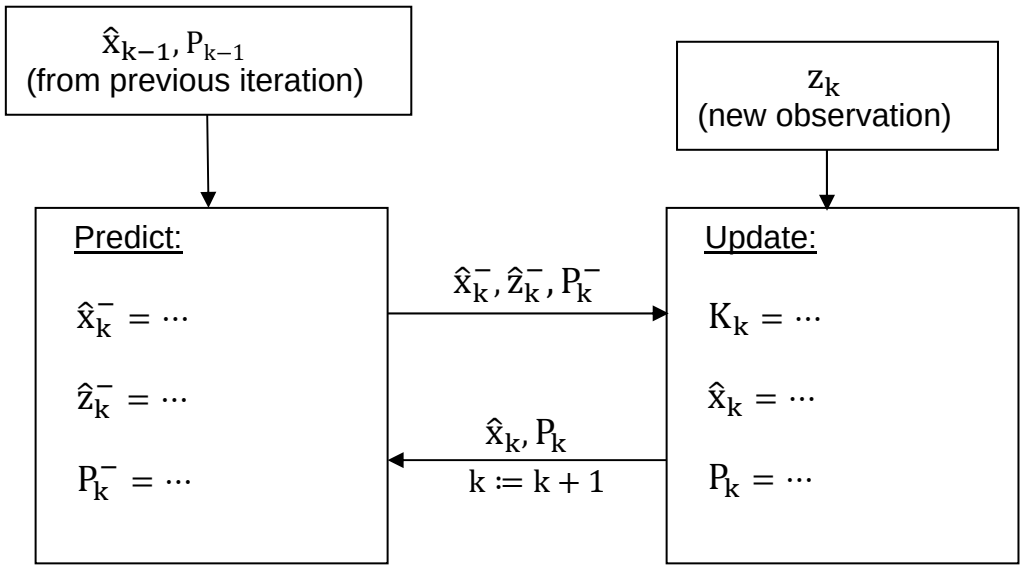
The signal and observation models for a non-linear system without control inputs are, respectively

$$x_k = f(x_{k-1}, w_{k-1}) = x_{k-1}^2 + w_{k-1}, \qquad E[w_k] = 0, \quad E[w_k^2] = \sigma_w^2$$
$$z_k = h(x_k, v_k) = x_k^3 + v_k, \qquad E[v_k] = 0, \quad E[v_k^2] = \sigma_v^2$$

Calculate the Jacobians

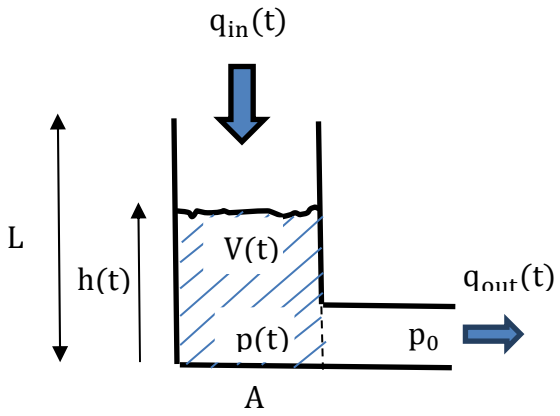
- $A_k = \left.\frac{\partial f}{\partial x}\right|_{\hat{x}_{k-1}}$
- $W_k = \left.\frac{\partial f}{\partial w}\right|_{\hat{x}_{k-1}}$
- $H_k = \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_k^-}$
- $V_k = \left.\frac{\partial h}{\partial v}\right|_{\hat{x}_k^-}$

Find the Extended Kalman filter equations for this system on the form



Problem 6.2 (Theoretical + MATLAB)
Water tank system:
Extended Kalman filter and non-linear system state estimation

Below is a system sketch of at water tank of height L and bottom area A. The tank that has an input flow $q_{in}(t)$ (control input) and an output flow $q_{out}(t)$. The output flow is determined by the difference between the liquid pressure at the bottom of the tank and the liquid pressure in the outflow channel.



The **system dynamics** can be described as below:

Input flow $q_{in}(t)$

Under certain conditions:

Output flow: $q_{out}(t) = K_v\sqrt{p(t) - p_0}$ where K_v depends on topologi and liquid

Pressure at bottom of tank due to gravity on liquid: $p(t) = \rho g h(t)$

Dynamics of liquid volume in tank: $\frac{dV}{dt} = q_{in}(t) - q_{out}(t) = A\frac{dh}{dt}$

Combining terms gives non-linear differential equation $\frac{dh}{dt} = \frac{q_{in}(t)}{A} - \frac{K_v}{A}\sqrt{\rho g h(t) - p_0}$

The system dynamics can be put into **continuous-time non-linear state space form:**

true state: $x(t) = h(t)$
observed state: $z(t)$
input: $u(t) = q_{in}(t)$
process noise: $w(t) \sim N(0, \sigma_w^2)$
obs. noise: $v(t) \sim N(0, \sigma_v^2)$

process model:	$\frac{dx}{dt} = -\frac{K_v}{A}\sqrt{\rho g x(t) - p_0} + \frac{u(t)}{A} + w(t)$
observation model:	$z(t) = x(t) + v(t)$

Theoretical part of problem:

Convert the continuous-time non-linear state space model to a discrete-time non-linear state space model on the form

$$x_k = f(x_{k-1}, u_{k-1}) + w_{k-1}, \qquad E[w_k] = 0, \quad E\left[w_k^2\right] = \sigma_w^2$$
$$z_k = h(x_k) + v_k, \qquad E[v_k] = 0, \quad E\left[v_k^2\right] = \sigma_v^2$$

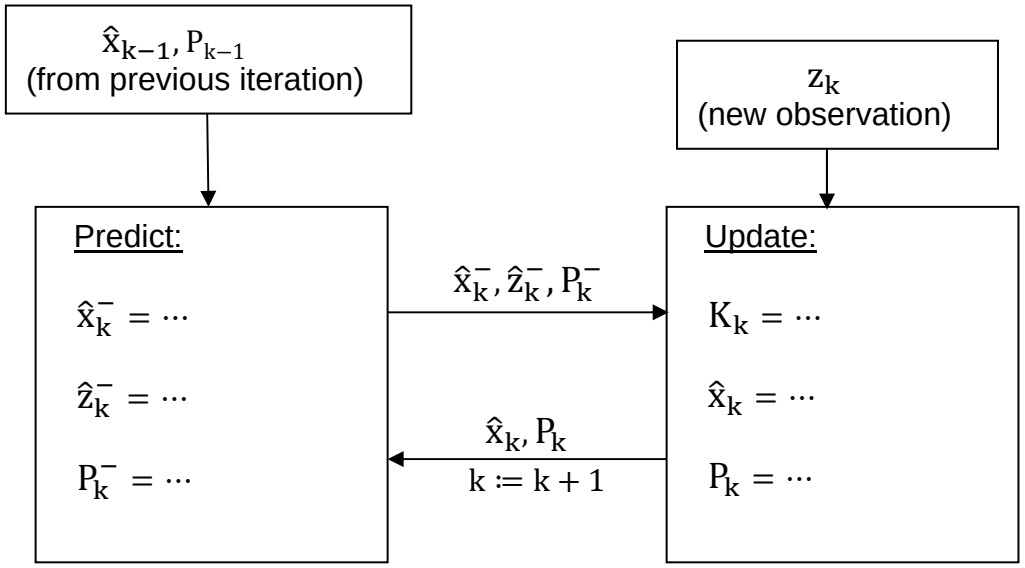
(Hint: use a simple first-order discrete-time approximation to continuous-time derivative:

$\frac{dr}{dt} = s(t) \Rightarrow r_k \cong r_{k-1} + s_{k-1}T_s$ where T_s is the sampling period)

Calculate the Jacobians

- $A_k = \left.\frac{\partial f}{\partial x}\right|_{\hat{x}_{k-1}}$
- $W_k = \left.\frac{\partial f}{\partial w}\right|_{\hat{x}_{k-1}}$
- $H_k = \left.\frac{\partial h}{\partial x}\right|_{\hat{x}_k^-}$
- $V_k = \left.\frac{\partial h}{\partial v}\right|_{\hat{x}_k^-}$

Thus, **find the Extended Kalman filter equations** for this system on the form



MATLAB part of problem:

Implement the non-linear system and an Extended Kalman filter for estimating water level as a function of time

You can, e.g., use the following system parameters (given without units)

$N = 100$
$f_s = 10$
$L = 1 \;\Rightarrow\; 0 \leq h(t) \leq 1$
$A = 1$
$\rho = 1000$
$g = 9.82$
$p_0 = 1000$
$K_v = 0.01$
$\sigma_w^2 = 10^{-4}$
$\sigma_v^2 = 10^{-2}$
$q_{in}(t) = q_{in} = 0.5$

Make plots of true, observed and estimated water levels
Plot estimation errors and calculate MSE's

Try different initializations such as half full tank, empty tank or full tank
Try to change q_{in} to higher and lower values.
Optionally implement also the standard Kalman filter and compare with the Extended Kalman filter with respect to water level estimation capability

Problem 6.3 (MATLAB)
Extended Kalman filter and non-linear system state estimation

The signal and observation model for a non-linear system without control inputs can be expressed as

$$x_k = f(x_{k-1}) + w_k, \qquad E[w_k] = 0, \quad E\left[w_k^2\right] = \sigma_w^2$$
$$z_k = h(x_k) + v_k, \qquad E[v_k] = 0, \quad E\left[v_k^2\right] = \sigma_v^2$$

where “f” and “h” are assumed to be known non-linear functions.

MATLAB implementation:

Define first “f” and “h” to be **linear** functions

- Define process and observation noise power
- Define process and observation noise distributions to be zero-mean gaussian
- Choose a signal length, e.g. N=100 or more
- Implement the signal model (including initialization)
- Implement the observation model
- Implement the standard Kalman filter algorithm (including initialization)
- Implement the extended Kalman filter algorithm (including initialization)
- Calculate MSE’s for observations and estimated states for KF and EKF
- Make plots of true states, observed states and estimated states for KF and EKF
- Make plots of Kalman gains and estimated MSE’s for KF and EKF
- Explain

Next, define “f” and “h” to be **non-linear** functions, repectively

- weakly non-linear
- moderately non-linear
- strongly non-linear

- Repeat the simulations from above

Then, allow **process and observation noise** to be **non-gaussian** (skewed), but still zero-mean.

- Repeat the simulations from above for both linear and non-linear “f” and “h”